15 Prove algebraically that the sum of the squares of any two consecutive even numbers is always a multiple of 4

(Total for Question 15 is 3 marks)

16 y is inversely proportional to the square of x.

$y = 8$ when $x = 2.5$

Find the negative value of x when $y = \frac{8}{9}$

$y = k/x^2$, so $8 = k/(2.5)^2$.

$k = 8 \times 6.25 = 50$.

$8/9 = 50/x^2$, so $x^2 = 50 \div (8/9) = 56.25$.

Take the negative root: $x = -7.5$.

3 marks total
Inverse-square form
k = 50
Negative x = −7.5

(Total for Question 16 is 3 marks)

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